Practice Set 8 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. What are the various problems in PSA screening

Type: LAQ (25 marks)

Introduction

- Screening tests healthy, symptom-free men; its aim is lower prostate cancer mortality through earlier treatment.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, mortality has fallen by >40% (SEER data), but at the price of overdiagnosis.

Evidence from screening trials

Trial	Design	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2–4 years vs none	**20% relative fall** in cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian), USA	Annual PSA vs usual care	**No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power
Göteborg-1 (Sweden)	Biennial PSA, men 50–64	44% fall in cancer death at 14 years
CAP (Cluster randomised trial of PSA testing for prostate cancer), UK	Single PSA invitation	No mortality difference at 10 years; only a small fall at 15 years

- Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years.

Problems in PSA screening

1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

Guideline positions

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2012	Grade D: against screening
USPSTF 2018 (current)	**Grade C, 55–69 years: individual shared decision; Grade D ≥70**
AUA (American Urological Association)	Shared decision 55–69; every 2 years
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted early detection; earlier start (45) with family history or African descent, 40 for BRCA2 carriers

- No body recommends unselected population screening.

Solutions: risk-adapted, MRI-first pathway

```
Informed man, life expectancy >10–15 y
   → PSA (repeat if raised)
   → risk stratify: PSA density, free/total PSA, risk calculator,
     PHI / 4Kscore / Stockholm3
   → MRI (magnetic resonance imaging)
        PI-RADS 1–2, low density → no biopsy, keep in follow-up
        PI-RADS ≥3 → targeted ± systematic biopsy (transperineal)
   → Grade Group 1 → active surveillance
```

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals.
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease uncouples overdiagnosis from overtreatment.

- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

Diagram to draw: iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart.

> **Key points to remember:**
> - ERSPC: 20% fall in cancer death; PLCO null from contamination.
> - Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
> - USPSTF 2018: Grade C at 55–69, Grade D ≥70.
> - Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1.

Q2. Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction.

Type: LAQ (25 marks) (principles also answered in Practice Set 3 Paper 4, Q2)

Assumed meaning: GI = gastrointestinal (bowel) reconstruction, i.e. antegrade continence enema channels.

Introduction

- A continent diversion empties at will: by voiding through the urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous**).
- **Continence is a pressure balance:** reservoir pressure must stay below outlet resistance. Failure of either side causes leakage.

Principles of continence

A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding break up coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** ($P = 2T/R$), a larger radius gives lower pressure for the same wall tension.
- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

B. Reliable outlet

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** an efferent channel that is easy to catheterise and has a durable continence mechanism.

Mechanism	Example	How it works	Weakness
Nipple valve (intussuscepted)	Kock, Mainz I	Reservoir pressure compresses the intussuscepted segment	Same pressure effaces it; failure ≥15%, reoperation about one-third
Flap valve	**Mitrofanoff**, Yang–Monti, Kropp	Narrow tube in a submucosal tunnel is coapted as pressure rises	No pop-off; stomal stenosis
Imbricated ileocaecal valve + tapered ileum	Indiana pouch	Continence zone lies at the plicated valve	Bowel frequency, B12 loss

- **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure.

C. Protect the upper tracts

- Unobstructed ureterointestinal anastomosis; eGFR (estimated glomerular filtration rate) about ≥35–40 mL/min for the acid load; ability to self-catheterise.

The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable**

channel.

- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

Conduits

Conduit	Points
Appendix (first choice)	Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length
Yang–Monti	2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions
Tapered ileum	Long enough; kinks if redundant
Ureter, fallopian tube	More stenosis
Casale bladder flap	No bowel; consumes bladder capacity

Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.
2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.
3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

Applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

Applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button.

Results and complications

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

Diagram to draw: flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

Key points to remember:

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Continence >95%; stomal stenosis is the commonest complication.

Q3a. Pre-op assessment of unrelated donor

Type: SAQ (10 marks) (related: 2023-Winter Paper 4, Q3e; 2022-Summer Paper 3, Q3a)

Assumed meaning: evaluation of a living unrelated kidney donor (Indian context).

Principles

- **Donor safety first**; the medical work-up equals that of a related donor.
- The extra step is **legal and ethical**: proof of genuine, unpaid motive.

Legal assessment (THOA: Transplantation of Human Organs Act 1994, amended 2011; Rules 2014)

- Donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

Medical assessment

1. **Immunological:** ABO group, HLA (human leukocyte antigen) typing, crossmatch, DSA (donor-specific antibody). More HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.
2. **Kidney function** by two methods, one measured (e.g. DTPA [diethylenetriamine pentaacetic acid] GFR [glomerular filtration rate]). KDIGO (Kidney Disease: Improving Global Outcomes) 2017: GFR ≥90 accept, <60 mL/min/1.73 m² decline; albuminuria >100 mg/g declines.
3. **Metabolic:** HbA1c (glycated haemoglobin), lipids, coagulation; diabetes, significant obesity and uncontrolled hypertension exclude.
4. **Infection:** HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), syphilis, tuberculosis.
5. **Cardiopulmonary** and age-appropriate **cancer screening**.
6. **Urological:** stones (bilateral exclude), haematuria, malignancy, recurrent UTI (urinary tract infection).
7. **Anatomy:** CT (computed tomography) angiography for vessels; split function if asymmetric; the better kidney stays with the donor.
8. **Psychosocial** evaluation; independent donor advocate.

Counselling

- Function settles at 60–80% of baseline; mortality 3 per 10,000; lifetime ESRD (end-stage renal disease) 90 vs 14 per 10,000 (Muzaale 2014).

Diagram to draw: flowchart: legal approval → compatibility → medical → anatomy → surgery.

Key points to remember:

- Unrelated donor = Authorisation Committee approval (THOA).
- Interview video-recorded; no commerce.
- KDIGO 2017: GFR ≥90 accept, <60 decline.
- CT angiography; better kidney stays with donor.

Q3b. Treatment of inferior calyceal calculus

Type: SAQ (10 marks)

Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

Treatment by size

Size	Options
Asymptomatic, small	Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational
<10 mm	SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either
10–20 mm	EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy)
>10 mm (AUA)	**PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line
>20 mm	PCNL first; staged RIRS an alternative

Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

Diagram to draw: infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

Key points to remember:

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

Q3c. Von Hippel Lindau disease

Type: SAQ (10 marks) (VHL gene also answered in Practice Set 2 Paper 2, Q3f)

Introduction

- VHL (von Hippel–Lindau) disease is an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma).

Genetics and pathogenesis

- **VHL tumour suppressor gene, chromosome 3p25** (1993). VHL protein forms an E3 ubiquitin ligase that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- Loss of VHL stabilises HIF (chiefly HIF-2α), which switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α and erythropoietin: hence **highly vascular** tumours.
- Knudson two-hit: germline plus somatic hit, so tumours are **early, bilateral and multifocal**.

Manifestations

Lesion	Frequency
Craniospinal haemangioblastoma (cerebellum commonest)	60–80%
Retinal angioma (earliest)	25–60%
Clear cell RCC (mean age 39)	25–75%
Renal cysts (commonest renal lesion)	Up to 76%
Phaeochromocytoma	10–25%
Pancreatic cysts / neuroendocrine tumours	45–75% / 11–17%
Epididymal cystadenoma	25–60% of men

- Also endolymphatic sac tumour.
- **RCC is now the commonest cause of death.**

Genotype–phenotype

Type	Phaeochromocytoma	RCC
1 (deletions, truncations)	Low	High
2A (missense)	High	Low
2B	High	High
2C	Phaeochromocytoma only	Absent

Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

Surveillance (VHL Alliance 2020)

- Eye examination from infancy; metanephrines yearly from 5; brain and spine MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

Urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm.

- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021; responses in about half of renal tumours.

Diagram to draw: VHL–HIF pathway in normoxia vs VHL loss.

> **Key points to remember:**
> - AD; VHL gene 3p25; HIF-2α → VEGF.
> - Type 2 = phaeochromocytoma.
> - 3 cm rule; nephron-sparing surgery.
> - Belzutifan approved 2021.

Q3d. Pelvic # assessment and management

Type: SAQ (10 marks) (lower tract trauma in pelvic fracture also answered in Practice Set 6 Paper 2, Q2)

Assumed meaning: urological assessment and management of lower urinary tract injury in pelvic fracture.

Introduction

- Pelvic fracture accompanies 83–95% of bladder ruptures; urethral injury occurs in about 10% of male pelvic fractures. Associated injuries cause the deaths.
- Mechanism: **EPR (extraperitoneal rupture)** from shearing or bone spicules; **PFUI (pelvic fracture urethral injury)** at the bulbomembranous junction.

Assessment

- **ATLS (Advanced Trauma Life Support)** resuscitation first; pelvic binder and stabilisation.
- **Clinical:** blood at meatus, inability to void, palpable bladder (urethra); gross haematuria, suprapubic pain (bladder). **Vaginal and rectal examination.**
- **RUG (retrograde urethrogram) before any catheter if blood at meatus**: 20 mL contrast via catheter tip in fossa navicularis, oblique position.
- **Cystography:** absolute indication **gross haematuria + pelvic fracture**. CT cystography: retrograde fill **350 mL** of 2–4% contrast (clamping after IV contrast is inadequate). Plain cystogram needs a drainage film.
- EPR: flame-shaped contrast; **IPR (intraperitoneal rupture)**: contrast outlining bowel.

Classification

- Bladder: AAST (American Association for the Surgery of Trauma) grades I–V.
- Urethra: Goldman I–V; practically partial vs complete disruption.

Management

Bladder

- Uncomplicated EPR: urethral catheter 2–3 weeks; cystogram before removal.
- **Immediate repair:** IPR, bladder neck injury, bone fragment in bladder, rectal or vaginal injury, open pelvic fixation or laparotomy.

Urethra

```
Blood at meatus → RUG
  Partial → one gentle catheter attempt
  Complete → suprapubic catheter (± primary realignment if stable)
  Bladder neck / rectal injury → immediate repair
Delayed (2–3 months) → up-and-down-o-gram → perineal excision
  and primary anastomosis (corporal split → pubectomy → rerouting)
```

Complications

- Stricture, ED (erectile dysfunction, mainly from the injury), incontinence (uncommon), fistula; missed IPR: ascites, azotaemia.

Diagram to draw: flame-shaped EPR vs IPR on cystogram; bulbomembranous distraction.

Key points to remember:

- Blood at meatus → RUG before catheter.
- Gross haematuria + pelvic fracture → CT cystogram 350 mL.
- EPR catheter; IPR repair.
- PFUI: SPC, delayed anastomotic urethroplasty.

Q3e. Brachytherapy for CA prostate

Type: SAQ (10 marks)

Definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT (external beam radiotherapy) can safely give. Prostate cancer's low α/β ratio also favours high dose per fraction (HDR).

Types

Feature	LDR (low-dose-rate), permanent seeds	HDR (high-dose-rate), temporary
Source	^{125}I (half-life 59.6 days), ^{103}Pd (17 days), ^{131}Cs (9.7 days)	**^{192}Ir** in remote afterloader
Monotherapy dose	^{125}I 140–160 Gy; ^{103}Pd 110–125 Gy	27 Gy/2, 31.5 Gy/3, 34–38 Gy/4 fractions
Dosimetry	Post-implant CT at 14–30 days	Real time; none afterwards

Technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.
- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 cm³) limits. Low D90 predicts failure.

Indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT gave no benefit, more toxicity).
- **Boost with EBRT 45 Gy ± ADT (androgen deprivation therapy):** unfavourable intermediate and high risk. **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, but grade 3 urinary toxicity 18.4% vs 5.2%.
- Also salvage and focal BT.

Contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP defect; gland >60 cm³ (cytoreduce with ADT); pubic arch interference; prior pelvic radiation; inflammatory bowel disease; cannot lie in lithotomy.

Complications

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).

Follow-up

- Nadir is deeper and later (about 2.5 years). **PSA bounce** in about one-third: observe for 30 months before biopsy. Failure: Phoenix (nadir + 2 ng/mL).

Diagram to draw: TRUS-template transperineal implant set-up.

Key points to remember:

- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

Q3f. Intravesical therapy for TCC of bladder

Type: SAQ (10 marks) (also answered in 2017-Summer Paper 4, Q1)

Introduction

- For NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) after complete TURBT (transurethral resection of bladder tumour). Problems: recurrence and progression.
- **Chemotherapy reduces recurrence; only BCG (bacille Calmette–Guérin) also reduces progression.**

Risk-adapted use

Risk	Therapy
Low (solitary low-grade Ta ≤3 cm)	Single post-op instillation
Intermediate	Chemotherapy or BCG induction + 1-year maintenance
High (HG [high-grade] T1, CIS, multifocal HG)	BCG induction + 3-year maintenance

1. Single immediate instillation

- Kills floating cells before implantation, so give **within 6 hours** (not at 24 hours). MMC (mitomycin C) 40 mg, epirubicin or **gemcitabine 2 g (SWOG S0337)**.
- Avoid after perforation. **Never BCG** immediately (sepsis).

2. Adjuvant chemotherapy

- MMC weekly ×6, optimised (empty bladder, fluid restriction, alkalinised urine). Gemcitabine; **Gem/Doce** (gemcitabine–docetaxel) for BCG shortage or unresponsive disease.

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin** → cytokines → **Th1 (T-helper 1)** response kills tumour.
- Start 2–4 weeks after TURBT; retain 1–2 hours.
- **SWOG schedule:** 6 weekly induction, then 3 weekly doses at 3, 6 months and every 6 months for 3 years. **NIMBUS:** reduced frequency inferior.
- Contraindications: gross haematuria, traumatic catheterisation, active UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis (symptomatic); persistent fever: isoniazid + rifampicin; sepsis (0.5%): stop BCG, triple therapy (not pyrazinamide) ± steroids.

BCG-unresponsive disease

- Recurrent or persistent HG disease despite **adequate BCG** (≥5 of 6 induction + ≥2 of 3 maintenance).
- **Timely radical cystectomy** is standard.
- Unfit or refusing: **pembrolizumab** (KEYNOTE-057; CIS complete response 41%), **nadofaragene firadenovec**, **N-803 + BCG**, Gem/Doce, TAR-200 device.

Device-assisted

- EMDA (electromotive drug administration) MMC; chemohyperthermia (RITE, HIVEC); UGN-102 gel.

Diagram to draw: BCG mechanism and SWOG schedule timeline.

Key points to remember:

- Single instillation within 6 hours; never BCG.
- Only maintenance BCG reduces progression.
- SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab 41% CR.

Rapid Revision Sheet – Practice Set 8 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Continence >95%; stomal stenosis is the commonest complication.

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- Unrelated donor = Authorisation Committee approval (THOA).
- Interview video-recorded; no commerce.
- KDIGO 2017: GFR ≥90 accept, <60 decline.
- CT angiography; better kidney stays with donor.

Q3b. Treatment of inferior calyceal calculus (SAQ)

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

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